

Optimal (Multi-Stage) Treatment Regimes

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PubH 7485/8485

Optimal Single Stage Treatment Regime

- Typically, to identify the optimal treatment strategy, we would posit a regression model $E(Y|A, X) = \mu(A, X; \eta)$ and estimate the parameter η using least squares or maximum likelihood estimators which we denote by $\hat{\eta}$
- In this case we could estimate the optimal treatment regime by $\hat{g}^{opt} = I(\mu(1, x; \hat{\eta}) \geq \mu(0, x; \hat{\eta}))$
- We have seen that there has been work to avoid this “two-step” procedure (OWL)

Optimal Multi-Stage Treatment Regime

- Previously, we have talked about estimating the anticipated (mean) response if everyone in the population were to follow a particular (dynamic) treatment regime.
- We could test a variety of different treatment strategies and pick the one which is optimal (i.e., has the highest estimated mean response)
- Instead I may want to find the optimal regime
- Can we extend the regression-approach when I want to estimate a multi-stage treatment strategy?

Optimal Multi-Stage Treatment Regime

First-pass approaches in the context of the HIV problem we have worked on for homework. Each of these is problematic. Why?

- Consider finding the optimal first stage decision. Fit a regression model for (log) CD4 count at 6 months with covariates at baseline (0 months), treatment decision at baseline, and their interaction. That is, $E(CD4_1|A_0, L_0) = \mu(A_0, L_0; \eta_0)$. Estimate the optimal treatment regime at baseline by $\hat{g}_0^{opt} = I(\mu(1, l_0; \hat{\eta}_0) \geq \mu(0, l_0; \hat{\eta}_0))$.
- Then for the optimal decision at 6 months, fit a regression model for (log) CD4 count at 12 months with covariates at baseline and 6 months, treatment decision at baseline and 6 months, and possible interactions. That is, $E(CD4_2|A_1, H_1) = \mu(A_1, H_1; \eta_1)$, where $H_1 = (L_0, A_0, L_1)$. Estimate the optimal treatment regime by $\hat{g}_1^{opt} = I(\mu(1, h_1; \hat{\eta}_1) \geq \mu(0, h_1; \hat{\eta}_1))$
- Continue on in this manner for the remaining decision points

Optimal Multi-Stage Treatment Regime

First-pass approaches in the context of the HIV problem we have worked on previously. Each of these is problematic. Why?

- Consider finding the optimal first stage decision. Fit a regression model for (log) CD4 count at 24 months (final outcome) with covariates at baseline (0 months), treatment decision at baseline, and their interaction. That is, $E(CD4_4|A_0, L_0) = \mu(A_0, L_0; \eta_0)$. Estimate the optimal treatment regime at baseline by $\hat{g}_0^{opt} = I(\mu(1, l_0; \hat{\eta}_0) \geq \mu(0, l_0; \hat{\eta}_0))$.
- Then for the optimal decision at 6 months, fit a regression model for (log) CD4 count at 24 months with covariates at baseline and 6 months, treatment decision at baseline and 6 months, and possible interactions. That is, $E(CD4_4|A_1, H_1) = \mu(A_1, H_1; \eta_1)$, where $H_1 = (L_0, A_0, L_1)$. Estimate the optimal treatment regime by $\hat{g}_1^{opt} = I(\mu(1, h_1; \hat{\eta}_1) \geq \mu(0, h_1; \hat{\eta}_1))$
- Continue on in this manner for the remaining decision points

Optimal Multi-Stage Treatment Regime

First-pass approaches in the context of the HIV problem we have worked on previously. Each of these is problematic. Why?

- Consider finding the optimal first stage decision. Fit a regression model for (log) CD4 count at 24 months (final outcome) with all covariate and treatment history recorded. That is,

$$E(CD4_4 | A_0, L_0, A_1, \dots, A_3, L_3) = \mu(A_0, L_0, A_1, \dots, A_3, L_3; \eta).$$

Estimate the optimal treatment regime at baseline by

$$\hat{g}_0^{opt} = I(\mu(1, l_0, a_1, \dots, a_3, l_3; \hat{\eta}) \geq \mu(0, l_0, a_1, \dots, a_3, l_3; \hat{\eta})).$$

- Then for the optimal decision at 6 months, using the same regression model as before, estimate the optimal treatment regime by
- $$\hat{g}_1^{opt} = I(\mu(a_0, l_0, 1, \dots, a_3, l_3; \hat{\eta}) \geq \mu(a_0, l_0, 0, \dots, a_3, l_3; \hat{\eta}_1))$$
- Continue on in this manner for the remaining decision points

Optimal Multi-Stage Treatment Regime

- The preceding examples should (hopefully) convince you that finding the optimal regime starting at the first time point and going forward is problematic
- Ideally, we would start at the last time point and find the optimal regime at that time point using regression-based techniques.
- Then if we knew (or could estimate) the response assuming all subjects followed the optimal regime at the last decision point, we could use that outcome in a regression model to estimate the optimal regime at the $M - 1$ decision point.
- We could continue using this recursive pattern until we reach the first decision point.

Notation: let $H_j = (L_0, A_0, \dots, L_j)$ denote the history (covariate and treatment) up until the j^{th} decision point

- ① Start at the last decision point (M^{th} time point). Finding the optimal regime at the last decision point is equivalent to a single stage decision point. Posit a regression model for the outcome (Y) given H_M and A_M (that is $Q_M(A_M, H_M) = E(Y|A_M, H_M) = \mu(A_M, H_M; \eta_M)$) and estimate the regression parameters. Estimate the optimal treatment regime at t_M by $\hat{g}_M^{opt} = I(\mu(1, h_M; \hat{\eta}_M) \geq \mu(0, h_M; \hat{\eta}_M))$.
- ② Estimate the anticipated response $V_M(H_M)$ if all subjects were to follow the optimal regime at time t_M . That is,
$$\hat{V}_M(H_M) = \max_{a_M} \mu(a_M, H_M; \hat{\eta}_M)$$

- ③) Posit a regression model for the outcome if all subjects were to follow the optimal regime at time t_M given H_{M-1} and A_{M-1} (that is $Q_{M-1}(A_{M-1}, H_{M-1}) = E(\hat{V}_M | A_{M-1}, H_{M-1}) = \mu(A_{M-1}, H_{M-1}; \eta_{M-1})$) and estimate the regression parameters. Estimate the optimal treatment regime at t_{M-1} by $g_{M-1}^{opt} = I(\mu(1, h_{M-1}; \hat{\eta}_{M-1}) \geq \mu(0, h_{M-1}; \hat{\eta}_{M-1}))$
- ④) Estimate the anticipated response $\hat{V}_{M-1}(H_{M-1})$ if all subjects were to follow the optimal regime at time t_M and t_{M-1} . That is, $\hat{V}_{M-1}(H_{M-1}) = \max_{a_{M-1}} \mu(a_{M-1}, H_{M-1}; \hat{\eta}_{M-1})$

- 5) Continue repeating the previous two steps until one reaches the decision point at time t_0 . That is, for the decision at time t_j , posit a regression model for the outcome if all subjects were to follow the optimal regime at all subsequent time points $(t_{j+1}, \dots, t_M)$ given H_j and A_j (that is $Q_j(A_j, H_j) = E(\hat{V}_j | A_j, H_j) = \mu(A_j, H_j; \eta_j)$) and estimate the regression parameters. Estimate the optimal treatment regime at t_j by $\hat{g}_j^{opt} = I(\mu(1, h_j; \hat{\eta}_j) \geq \mu(0, h_j; \hat{\eta}_j))$. Then estimate the anticipated response $\hat{V}_j(H_j)$ if all subjects were to follow the optimal regime at time t_j through t_M . That is, $\hat{V}_j(H_j) = \max_{a_j} \mu(a_j, H_j; \hat{\eta}_j)$

Estimated response under the optimal regimes

- Note that we could estimate $E(Y^{g^{opt}})$ by $n^{-1} \sum \hat{V}_0(H_0)$ (that is the average of the anticipated response if all subjects were to follow the optimal regime from time t_0 onward)
- This estimate will be upwardly biased
- We could also estimate $E(Y^{g^{opt}})$ using the other approaches that we have learned but this would require fitting additional models

HIV Example

- Start by fitting model for (log) CD4 at the final timepoint
- We will fit separate models for those who are resistant vs. those not
- Use backward variable selection

HIV Example - 18 month treatment decision

- Model among those who are not resistant at 18 months

```
m3_R30 <- lm(logCD44 ~ logCD43 + cumA2 + A3 +  
              (logCD43 + cumA2)*A3,  
              data = dataHW3[which(dataHW3$R3 == 0), ])  
round(summary(m3_R30)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-0.13	0.25	-0.52	0.60
## logCD43	0.98	0.04	23.63	0.00
## cumA2	0.00	0.03	-0.08	0.94
## A3	1.48	0.32	4.59	0.00
## logCD43:A3	-0.18	0.05	-3.34	0.00
## cumA2:A3	-0.10	0.04	-2.55	0.01

HIV Example - 18 month treatment decision

- Model among those who are resistant at 18 months

```
m3_R31 <- lm(logCD44 ~ logCD43 + cumA2 + A3 +  
             (logCD43 + cumA2)*A3,  
             data = dataHW3[which(dataHW3$R3 == 1), ])  
round(summary(m3_R31)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-0.49	0.20	-2.42	0.02
## logCD43	0.97	0.03	27.70	0.00
## cumA2	0.09	0.04	2.36	0.02
## A3	-0.13	0.33	-0.40	0.69
## logCD43:A3	0.07	0.06	1.24	0.22
## cumA2:A3	-0.12	0.07	-1.76	0.08

HIV Example - 18 month treatment decision

- Get the optimal treatment for each individual

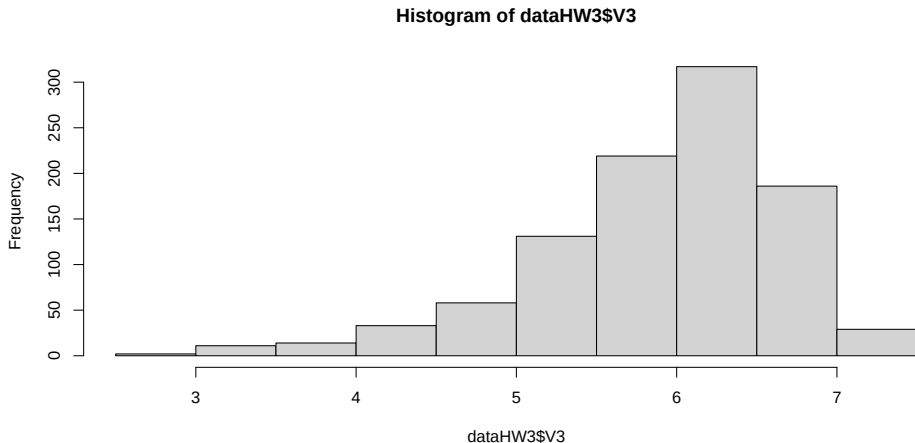
```
dataHW3_A31 <- dataHW3_A30 <- dataHW3
dataHW3_A31$A3 <- 1; dataHW3_A30$A3 <- 0
predA30 <- predA31 <- NA
R31 <- which(dataHW3_A30$R3 == 1)
R30 <- which(dataHW3_A30$R3 == 0)
predA30[R31] <- predict(m3_R31, newdata = dataHW3_A30[R31, ])
predA31[R31] <- predict(m3_R31, newdata = dataHW3_A31[R31, ])
predA30[R30] <- predict(m3_R30, newdata = dataHW3_A30[R30, ])
predA31[R30] <- predict(m3_R30, newdata = dataHW3_A31[R30, ])
gopt3 <- ifelse(predA31 > predA30, 1, 0)
table(gopt3)
```

```
## gopt3
##      0      1
## 137 863
```

HIV Example - 18 month treatment decision

- Estimate the value function

```
dataHW3$V3 <- pmax(predA31, predA30)  
hist(dataHW3$V3)
```



HIV Example - 12 month treatment decision

- Model among those who are not resistant at 12 months

```
m2_R20 <- lm(V3 ~ logCD42 + cumA1 + A2 +  
              (logCD42 + cumA1)*A2,  
              data = dataHW3[which(dataHW3$R2 == 0), ])  
round(summary(m2_R20)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	1.24	0.22	5.61	0.00
## logCD42	0.80	0.04	21.98	0.00
## cumA1	-0.16	0.03	-5.22	0.00
## A2	1.05	0.30	3.45	0.00
## logCD42:A2	-0.15	0.05	-2.92	0.00
## cumA1:A2	-0.05	0.04	-1.22	0.22

HIV Example - 12 month treatment decision

- Model among those who are resistant at 12 months

```
m2_R21 <- lm(V3 ~ logCD42 + cumA1 + A2 +  
             (logCD42 + cumA1)*A2,  
             data = dataHW3[which(dataHW3$R2 == 1), ])  
round(summary(m2_R21)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-0.18	0.35	-0.51	0.61
## logCD42	0.89	0.06	15.17	0.00
## cumA1	-0.09	0.08	-1.08	0.28
## A2	-0.76	0.58	-1.30	0.19
## logCD42:A2	0.09	0.10	0.99	0.32
## cumA1:A2	0.20	0.14	1.41	0.16

HIV Example - 12 month treatment decision

- Get the optimal treatment for each individual

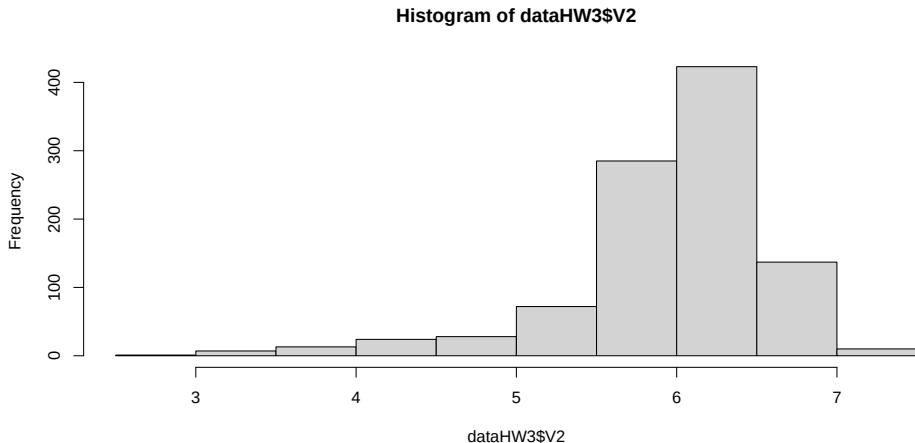
```
dataHW3_A21 <- dataHW3_A20 <- dataHW3
dataHW3_A21$A2 <- 1; dataHW3_A20$A2 <- 0
predA20 <- predA21 <- NA
R21 <- which(dataHW3$R2 == 1)
R20 <- which(dataHW3$R2 == 0)
predA20[R21] <- predict(m2_R21, newdata = dataHW3_A20[R21, ])
predA21[R21] <- predict(m2_R21, newdata = dataHW3_A21[R21, ])
predA20[R20] <- predict(m2_R20, newdata = dataHW3_A20[R20, ])
predA21[R20] <- predict(m2_R20, newdata = dataHW3_A21[R20, ])
gopt2 <- ifelse(predA21 > predA20, 1, 0)
table(gopt2)
```

```
## gopt2
##    0    1
## 227 773
```

HIV Example - 12 month treatment decision

- Estimate the value function

```
dataHW3$V2 <- pmax(predA21, predA20)  
hist(dataHW3$V2)
```



HIV Example - 6 month treatment decision

- Model among those who are not resistant at 6 months

```
m1_R10 <- lm(V2 ~ logCD41 + A0 + A1 +  
             (logCD41 + A0)*A1,  
             data = dataHW3[which(dataHW3$R1 == 0), ])  
round(summary(m1_R10)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	2.26	0.25	8.99	0.00
## logCD41	0.64	0.04	15.64	0.00
## A0	-0.30	0.04	-7.51	0.00
## A1	0.50	0.36	1.40	0.16
## logCD41:A1	-0.09	0.06	-1.54	0.13
## A0:A1	-0.01	0.06	-0.26	0.80

HIV Example - 6 month treatment decision

- Model among those who are resistant at 6 months

```
m1_R11 <- lm(V2 ~ logCD41 + A1 +  
             (logCD41)*A1,  
             data = dataHW3[which(dataHW3$R1 == 1), ])  
round(summary(m1_R11)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-1.47	0.77	-1.91	0.06
## logCD41	1.03	0.14	7.61	0.00
## A1	0.50	1.16	0.43	0.67
## logCD41:A1	-0.10	0.21	-0.49	0.63

HIV Example - 6 month treatment decision

- Get the optimal treatment for each individual

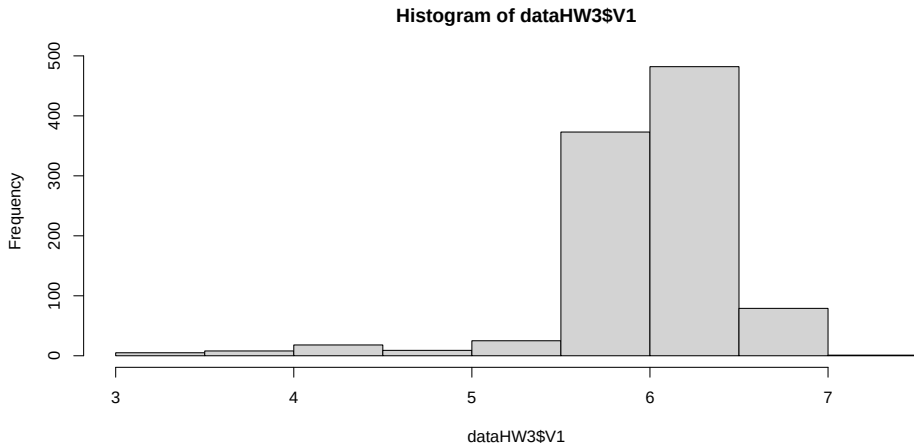
```
dataHW3_A11 <- dataHW3_A10 <- dataHW3
dataHW3_A11$A1 <- 1; dataHW3_A10$A1 <- 0
predA10 <- predA11 <- NA
R11 <- which(dataHW3$R1 == 1)
R10 <- which(dataHW3$R1 == 0)
predA10[R11] <- predict(m1_R11, newdata = dataHW3_A10[R11, ])
predA11[R11] <- predict(m1_R11, newdata = dataHW3_A11[R11, ])
predA10[R10] <- predict(m1_R10, newdata = dataHW3_A10[R10, ])
predA11[R10] <- predict(m1_R10, newdata = dataHW3_A11[R10, ])
gopt1 <- ifelse(predA11 > predA10, 1, 0)
table(gopt1)
```

```
## gopt1
##      0      1
## 913  87
```

HIV Example - 6 month treatment decision

- Estimate the value function

```
dataHW3$V1 <- pmax(predA11, predA10)  
hist(dataHW3$V1)
```



HIV Example - 0 month treatment decision

- Model among those who are not resistant at 6 months

```
m0 <- lm(V1 ~ logCD40 + A0 +  
          (logCD40)*A0,  
          data = dataHW3)  
round(summary(m0)$coefficients, digits = 2)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	2.32	0.38	6.12	0.00
## logCD40	0.62	0.06	10.11	0.00
## A0	0.27	0.50	0.54	0.59
## logCD40:A0	-0.08	0.08	-1.00	0.32

HIV Example - 0 month treatment decision

- Get the optimal treatment for each individual

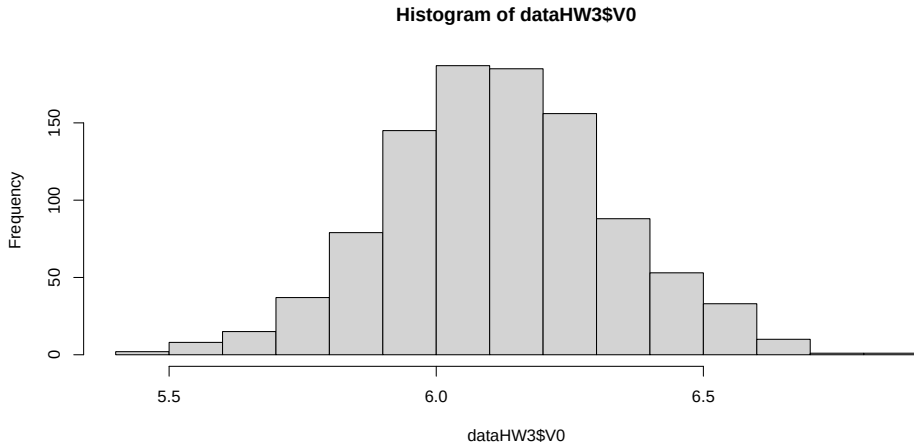
```
dataHW3_A01 <- dataHW3_A00 <- dataHW3
dataHW3_A01$A0 <- 1; dataHW3_A00$A0 <- 0
predA00 <- predA01 <- NA
predA00 <- predict(m0, newdata = dataHW3_A00)
predA01 <- predict(m0, newdata = dataHW3_A01)
gopt0 <- ifelse(predA01 > predA00, 1, 0)
table(gopt0)
```

```
## gopt0
##      0
## 1000
```

HIV Example - 0 month treatment decision

- Estimate the value function

```
dataHW3$V0 <- pmax(predA01, predA00)  
hist(dataHW3$V0)
```



HIV Example

- Potential improvement for derived DTR

```
mean(dataHW3$V0)
```

```
## [1] 6.118075
```

```
mean(dataHW3$logCD44)
```

```
## [1] 5.831221
```

Estimation of Dynamic Treatment Regimes with IPW

- Here we can obtain an estimate of the anticipated response for any decision rule involving as few of covariates as we would like
- Adjustment for confounding is taken care of using the weights
- Challenge of IPW estimators is that they tend to be highly variable (large standard errors).
- In particular very similar decision rules can lead to very different estimates of the anticipated response. Why?

Optimal Treatment Regimes within a (simple) Class

- Suppose we wanted to find the optimal rule for the HIV problem of the form “give treatment if CD4 count is less than x and the patient is not resistant”
- That is, we want to find the optimal x
- We could choose a fine grid for x (i.e., 200, 210, . . . , 600) and estimate the anticipated response under each x and pick the maximum
- BUT - very similar decision rules can lead to very different estimates of the anticipated response.
- How could we fix that problem?

Dynamic Marginal Structural Models

- 1) Ensure that the decision rule we want to estimate can be indexed by some value x (or several values if relevant)
- 2) Create a grid of x values over the range of values that we wish to estimate, e.g. $(x_1, \dots, x_p)$
- 3) For x_1 , obtain the inverse probability weights to estimate that regime of interest
- 4) Create a dataset with the response variable (Y), the weights just calculated in step 3 (call this W), and a covariate (call this X) which is equal to x_1 for all individuals in the dataset

Note on Step 3

- The denominator of the weights is the probability that a particular subject would have followed the treatment regime of interest given their covariate history
- The numerator of the weights is an indicator for whether or not a subject's observed treatment history is consistent (or compliant with) the treatment regime of interest
- We have shown that (when combined with the numerator) the denominator can be rewritten as the probability that a particular subject would have followed their observed treatment history. That is,

$$\frac{1}{\prod_{j=0}^M P_{A_j|\bar{A}_{j-1}, \bar{L}_j}(A_{ij}|\bar{A}_{i,j-1}, \bar{L}_{ij})}$$

Dynamic Marginal Structural Models

- 5) For x_2 , obtain the inverse probability weights to estimate that regime of interest
- 6) Create a dataset with the response variable (Y), the weights just calculated in step 5 (call this W), and a covariate (call this X) which is equal to x_2 for all individuals in the dataset
- 7) Combine this dataset with the one in step 4
- 8) Repeat 5-7 for $x_3, \dots, x_p$

Dynamic Marginal Structural Models

- 9) With the combined dataset, fit regression model with Y as the outcome, X (and functions thereof) as the predictor, and W as the weight. Want to fit a flexible model (e.g., with quadratic or higher order polynomial terms)
- 10) Find the value of x which maximizes the anticipated response from this model. For example, suppose I fit a quadratic model $\beta_0 + \beta_1 X + \beta_2 X^2$. Which value of X gives the maximum? (CALCULUS!)

HIV Example

- Consider the following dynamic treatment regime: “Give treatment if the current CD4 count falls below x as long as the patient has not become resistant; otherwise do not treat.”
- Want to understand how the average CD4 count at 24 months would vary across different values of x

HIV Example

- Start by fitting model for probability of receiving treatment

##	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	2.96	0.38	7.80	0
## logCD4	-0.46	0.06	-7.50	0
## R	-1.10	0.12	-9.25	0

Denominator Weights (Part 1)

- calculate the probability of receiving treatment (i.e. $A = 1$) and
- probability the participant received treatment they actually did

```
dataHW3_long <- mutate(dataHW3_long,  
  pred_trt = predict(m3, newdata = dataHW3_long, type = "res  
  pred_trt_received = (pred_trt^A)*(1-pred_trt)^(1-A))
```

Denominator Weights (Part 2)

- calculate denominator weights

```
dataHW3$den_wt <- tapply(dataHW3_long$pred_trt_received,  
  dataHW3_long$ID, prod, na.rm = TRUE)
```

Numerator Weights

- Sample of data, let's assume that $x = 200$, which subject below are compliant with the regime of interest

##		R0	CD40	A0	R1	CD41	A1	R2	CD42	A2	R3	CD43	A3
##	1:	0	612	0	0	200	1	0	263	0	0	232	0
##	2:	0	367	0	0	354	1	0	547	1	0	634	1
##	3:	0	340	0	0	601	0	0	454	0	0	480	0
##	4:	0	579	0	0	373	0	0	262	0	0	216	0
##	5:	0	672	0	0	584	0	0	411	0	0	250	0
##	6:	0	455	0	0	332	1	0	436	1	0	357	1
##	7:	0	639	1	0	709	0	0	395	1	0	349	1
##	8:	0	710	0	0	878	1	0	1102	1	0	774	1
##	9:	0	481	0	0	236	0	0	206	0	0	417	0
##	10:	0	557	0	0	903	0	0	1464	0	0	1046	1

Numerator Weights

- Program the weights in R

```
threshold <- 200
dataHW3 <- mutate(dataHW3,
  # find which treatment a subject was supposed to receive
  # based on the regime
  A0_regime = ifelse(CD40 < threshold & R0 == 0, 1, 0),
  A1_regime = ifelse(CD41 < threshold & R1 == 0, 1, 0),
  A2_regime = ifelse(CD42 < threshold & R2 == 0, 1, 0),
  A3_regime = ifelse(CD43 < threshold & R3 == 0, 1, 0),
  # find out whether or not they were compliant
  compliant0 = ifelse(A0 == A0_regime, 1, 0),
  compliant1 = ifelse(A1 == A1_regime, 1, 0),
  compliant2 = ifelse(A2 == A2_regime, 1, 0),
  compliant3 = ifelse(A3 == A3_regime, 1, 0),
  # get numerator weight
  num_wt = compliant0*compliant1*compliant2*compliant3,
  # overall weight
```

Get IPW Estimate of Anticipated Response

- Weighted mean on both the original and log scale

```
weighted.mean(dataHW3$logCD44, dataHW3$wt)
```

```
## [1] 5.682566
```

```
weighted.mean(dataHW3$CD44, dataHW3$wt)
```

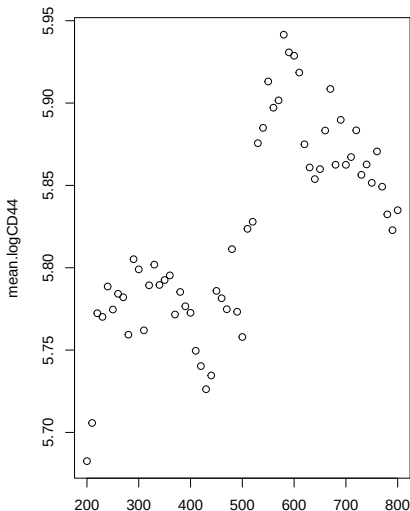
```
## [1] 349.4666
```

Consider a sequence of different x 's

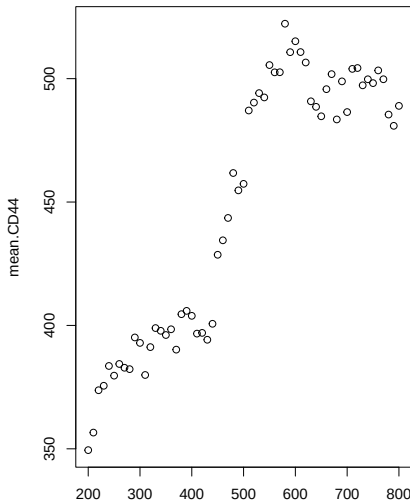
- Repeat the same procedure for $x = 200, 210, 220, \dots, 800$

Mean log CD4 at 24 mo.

Mean CD4 at 24 mo.



CD4 Threshold



CD4 Threshold

Dynamic Marginal Structural Model

- Intuitively may want to “smooth out” these plots
- One way to do this would be to have the estimated mean response be the outcome and threshold be the predictor (with possible nonlinear terms) in a regression model
- Such an approach ignores the fact that the precision of the estimated mean varies across different thresholds
- A more sophisticated approach: For each threshold, create a dataset with the response, IP weights, and threshold (same value for all subjects). Stack datasets across different thresholds and fit weighted regression model with threshold (and nonlinear terms) as predictor

Dynamic Marginal Structural Model

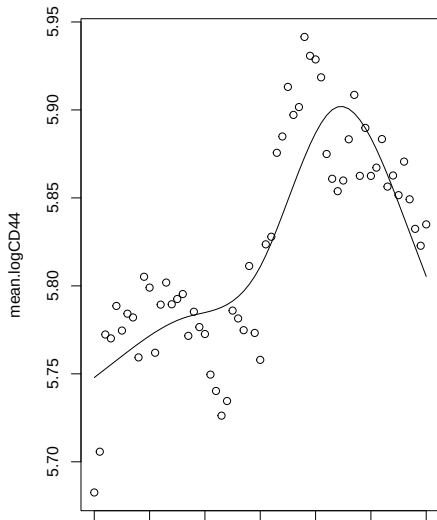
- Include restricted cubic spline basis in model. Output for model of mean CD44 shown

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	361.58	50.50	7.16	0.00
## rcs(threshold, 5)threshold	0.05	0.18	0.30	0.76
## rcs(threshold, 5)threshold'	1.70	1.11	1.53	0.13
## rcs(threshold, 5)threshold''	-4.13	3.22	-1.28	0.20
## rcs(threshold, 5)threshold'''	1.93	4.24	0.46	0.65

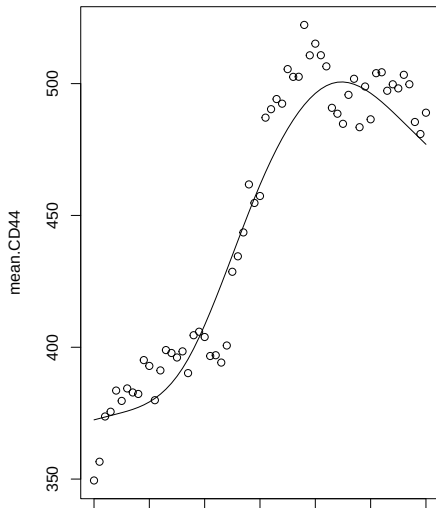
Dynamic Marginal Structural Model

- Pointwise IPW estimators compared with dynamic MSM

Mean log CD4 at 24 mo.



Mean CD4 at 24 mo.



Standard Errors

- Note that subjects can be consistent with more than one regime
- Subjects appear more than once in dataset with nonzero weights → correlated data
- Bootstrap can account for this, need to resample individuals not rows of data (when in long format)